



To: Dr. Steven Mitchell

Topic: HW5 Grade

From: Jaegon Hibbits

I believe my lane detection implementation for HW5 was marked down inconsiderably. This paper will argue why the homework submission proves more complete than the original grade.

[Original submission](#)

Original Working videos:

HW5-001.mp4

HW5-002.mp4

Original Non-functional Videos:

30.mp4

Test_video_01.mp4

Test_video_02.mp4

Changes:

The only changes implemented to have a decently working system were my parameters: aa crop value, cc crop value, Canny low, Canny high, Hough threshold, Min line length, Max line gap, Main display scale, Debug display scale, yellow_lower HSV, yellow_upper HSV, white_lower HSV, white_upper HSV, white_hls_lightness_min, white_hls_sat_max, White_gray_min. These changes themselves prove the script to be generally functional, as inconsistencies with the submission were purely due to filtering (typically under) and no cropping appropriately; The default crop is 600 which was applicable to the two working videos - HW5-001/002.mp4 - however, these were significantly larger videos than the other three which is the reason literally no results were able to be seen. Please view figures 1-3 to see the output from purely parameter changes.

[Figure 1](#) - 30.mp4

[Figure 2](#) - Test_video_01.mp4

[Figure 3](#) - Test_video_02.mp4

Code comparison video along with figure1 run. This is proof I was running the same code

[Comparison_run](#)



Conclusion:

Specifically, parameters had to be tweaked a significant amount to account for different lighting and the smaller video size. For every video, reducing the crop value to ~230 worked fantastic, this outputted the proper size/window for edge detection. From here, in 30.mp4, the yellow HSV thresholds were tightened to account for false detections from white-ish road edges and surrounding bright regions, while the white thresholds were slightly loosened because the dashed white lane markings were disappearing when the filter was too strict. In test_video_02.mp4, which was also a brighter video, the white thresholds had to be made more selective because the script was detecting road glare, shoulder edges, and other bright surfaces as possible lane lines. For this case, the white gray minimum and HLS lightness minimum were increased, the saturation limits were lowered, and the Hough settings were made stricter to prevent random edge fragments from becoming false lane lines. In test_video_01.mp4, which was the darker sunset video, the opposite adjustment was needed. The white thresholds, Canny thresholds, and Hough settings had to be loosened because the lane markings were dimmer and produced weaker edges. This proves the main issue was that the same parameter values could not work across all videos because lighting, contrast, video size, and lane visibility changed significantly between cases. With decently quick modifications ~45 minutes of work, I was able to determine working parameters, proving a working script.

Request:

If possible, I am requesting points back on this section of the HW; I received a 20/40 in this section with a working script, purely an error due to human incompetence and failure to find ideal parameters. Thank you.